Lecture 4: Determinants & Linear Transformations

(Ch. 4) (Ch. 6)

Determinants

$\det A$, $|A|$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\det A = a_{11} \cdot a_{22} - a_{12} \cdot a_{21} \neq 0$$

then A invertible

Note: $\det A \neq 0 \Leftrightarrow$ A invertible
(holds in general, not just 2x2)

Determinant for general (only square) matrix?

3 x 3: $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Submatrix A_{ij}: delete row i & column j from A

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$$

Example:

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 3 \\ 5 & 4 & 6 \end{bmatrix}$$

$$\det A = 2 \cdot \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} - 1 \cdot \begin{vmatrix} 4 & 3 \\ 5 & 6 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 2 \\ 5 & 4 \end{vmatrix}$$

$$= 2\cdot(2\cdot 6 - 3\cdot 4) - 1\cdot(4\cdot 6 - 3\cdot 5) + 3\cdot(4\cdot 4 - 2\cdot 5)$$
$$= 2\cdot(12-12) - 1\cdot(24-15) + 3(16-10)$$
$$= 2\cdot 0 - 1\cdot 9 + 3\cdot 6 = 9$$

Laplace's expansion theorem (Th. 4.5)

$$\det A = \sum_{k=1}^{n} (-1)^{i+k} a_{ik}\cdot \det A_{ik} \quad \text{(row } i\text{)}$$
$$= \sum_{k=1}^{n} (-1)^{k+j} a_{kj}\cdot \det A_{kj} \quad \text{(col } j\text{)}$$

(-1) part chess board

$$\begin{bmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & \cdots & \\ - & + & \cdots & & \\ + & \cdots & & & \\ \vdots & & & & \end{bmatrix}$$

Example

$$\det A = \begin{vmatrix} 2 & 1 & 3 \\ 4 & 2 & 3 \\ 5 & 4 & 6 \end{vmatrix} = -1\begin{vmatrix} 4 & 3 \\ 5 & 6 \end{vmatrix} + 2\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix}$$
$$= -1(4\cdot 6 - 3\cdot 5) + 2(2\cdot 6 - 3\cdot 5) - 4(2\cdot 3 - 3\cdot 4)$$
$$= -1(24-15) + 2(12-15) - 4(6-12)$$
$$= -1\cdot 9 + 2(-3) - 4(-6) = -9 - 6 + 24 = 9$$

Important Message	We can calculate the determinant more efficiently.

Tools

① If A is (upper or lower) triangular
$$\det A = a_{11}\cdot a_{22}\cdot a_{33}\cdot \cdots \cdot a_{nn}$$

② Elementary operations: A (original) $\to \tilde{A}$ (new)

- replacement (add a mult. of a row to another row)

$$\det \tilde{A} = \det A$$

- interchange two rows: $\det \tilde{A} = -\det A$
- scale a row by k: $\det \tilde{A} = k \cdot \det A$

③ $\det A = \det A^T$ (so we can also use column operations)

Example

$$\begin{vmatrix} 2 & 1 & 3 \\ 4 & 2 & 3 \\ 5 & 4 & 6 \end{vmatrix} \overset{\begin{matrix}-2\\-4\end{matrix}}{=} \begin{vmatrix} 2 & 1 & 3 \\ 0 & 0 & -3 \\ -3 & 0 & -6 \end{vmatrix} = -(-3)\begin{vmatrix} 2 & 1 \\ -3 & 0 \end{vmatrix}$$

$$= 3 \cdot (2 \cdot 0 - 1 \cdot (-3)) = 9$$

Question

a) $\det \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{bmatrix}$ b) $\det \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 5 & 5 \end{bmatrix}$ c) $\det \begin{bmatrix} 5 & -3 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 3 \end{bmatrix}$

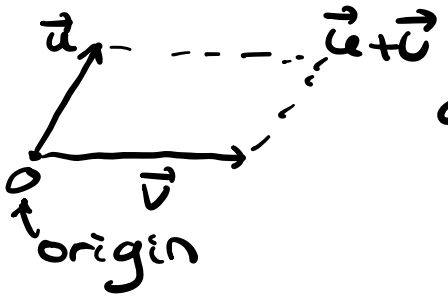
(a) upper triangular so $\det A = 1 \cdot 1 \cdot 5 = 5$

(b) we can turn the third row into zeros, so $\det A = 0$

(c) Matrix in (a) is echelon form of the matrix in (c) so $\det A = 5$

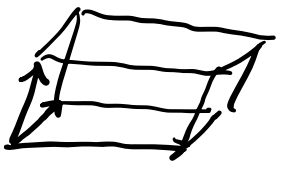
Properties

- Parallelogram



$\text{area} = |\det(\vec{u}\ \vec{v})|$

- Parallelepipid analogues



- $\det(A \cdot B) = \det(A) \cdot \det(B)$
- A is invertible $\Leftrightarrow \det A \neq 0$

Linear Transformations (Ch. 6)

Example $T: \mathbb{R}^2 \to \mathbb{R}^3$ (compare to calculus $f: \mathbb{R} \to \mathbb{R}$)

$$T(\vec{x}) = A \cdot \vec{x}$$

↑ representation matrix of T

$$\vec{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \qquad T(\vec{u}) = A \cdot \vec{u} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix}$$

T-image of vector $\vec{u}$

$$\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad T(\vec{v}) = A \cdot \vec{v} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}$$

- $T(\vec{u} + \vec{v}) = T\left(\begin{bmatrix} 3 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \\ 3 \end{bmatrix} = T(\vec{u}) + T(\vec{v})$
- $T(10 \cdot \vec{u}) = T\left(\begin{bmatrix} 20 \\ 10 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 10 \end{bmatrix} = \begin{bmatrix} 20 \\ 50 \\ 10 \end{bmatrix} = 10 \cdot T(\vec{u})$

Definition: Linear Transformation $T: \mathbb{R}^n \to \mathbb{R}^k$

(i) $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$

(ii) $T(c \cdot \vec{u}) = c \cdot T(\vec{u})$

} for all $\vec{u}, \vec{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$

Example: $T: \mathbb{R}^2 \to \mathbb{R}^3$ linear transformation

(in $\mathbb{R}^2$) $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ (in $\mathbb{R}^3$ $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$)

If $T(\vec{e}_1) = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $T(\vec{e}_2) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ then what is $T(\vec{x})$ for $\vec{x} \in \mathbb{R}^2$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad T(\vec{x}) = ?$$

$$T(\vec{x}) = T(x_1 \cdot \vec{e}_1 + x_2 \vec{e}_2) = T(x_1\vec{e}_1) + T(x_2\vec{e}_2)$$

$$= x_1 \cdot T(\vec{e}_1) + x_2 \cdot T(\vec{e}_2) = x_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$= \underbrace{\left(T(\vec{e}_1) \;\; T(\vec{e}_2)\right)}_{\text{representation matrix of } T} \cdot \vec{x}$$

$$A \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Important Message	Linear Transformations are matrix-vector multiplications

Note: The representation matrix of a linear transformation T is unique.

Questions: (i) Does any $\vec{y} \in \mathbb{R}^3$ have an $\vec{x} \in \mathbb{R}^2$ such that $T(\vec{x}) = \vec{y}$?

(ii) Does any image have a unique original?

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Answers:

(i) $\vec{y} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ There is no $\vec{x} \in \mathbb{R}^2$ such that $T(\vec{x}) = \vec{y}$

expl. $\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$x_1 = 1$

$2x_1 + x_2 = 0$

$x_2 = 1$

(T is not onto)

(ii) (For this T) yes Assume $T(\vec{u}) = T(\vec{v}) \Leftrightarrow$

$A \cdot \vec{u} = A \cdot \vec{v} \Leftrightarrow A^{-1}A\vec{u} = A^{-1}A\vec{v} \Leftrightarrow \vec{u} = \vec{v}$

(T is one-to-one) ← This is incorrect. We cannot use the inverse here because A is not a square matrix. (see very bottom for correct)

Properties of linear transformations

$T: \mathbb{R}^n \to \mathbb{R}^k$

onto: $\text{im } T = \mathbb{R}^k$ $\left(\text{Im } T = \{\vec{y} \in \mathbb{R}^k \mid T(\vec{x}) = \vec{y} \text{ for some } \vec{x} \in \mathbb{R}^n\}\right.$

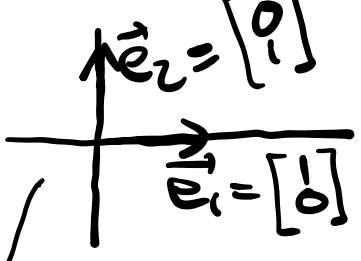
one-to-one: $\ker T = \{\vec{x} : T(\vec{x}) = \vec{0}\} = \{\vec{0}\}$

$(T(\vec{u}) = T(\vec{v}) \Rightarrow \vec{u} = \vec{v})$

$\Leftrightarrow$ onto: $\text{col } A = \mathbb{R}^k$

one-to-one: $\text{Null } A = \{\vec{0}\}$

Example T: rotation clockwise by 90° $\mathbb{R}^2 \to \mathbb{R}^2$



$T(\vec{e}_1) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$

$T(\vec{e}_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

repr. matrix $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$T: \mathbb{R}^n \to \mathbb{R}^n$ is <u>invertible</u> $\Leftrightarrow$ A is invertible

(we use T^{-1} for inverse transformation and this has repr. A^{-1})

T^{-1} would be rotating counter clockwise by 90°

$T^{-1}(\vec{x}) = A^{-1}(\vec{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \vec{x}$

$T^{-1}(T(\vec{x})) = \vec{x}$

<u>Composition:</u> See lecture notes.

Correction of (ii)

(ii) (For this T) yes. Assume $T(\vec{u}) = T(\vec{v}) \Leftrightarrow$

$A \cdot \vec{u} = A \cdot \vec{v} \Leftrightarrow A \cdot (\vec{u} - \vec{v}) = 0 \overset{(*)}{\Leftrightarrow} \vec{u} - \vec{v} = 0$

$\Leftrightarrow \vec{u} = \vec{v}$

(*) This holds if and only if $\ker T = \text{Null } A = \{\vec{0}\}$

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

So indeed, Null $A = \{\vec{0}\}$